

Programming HW 2

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Question 1

A	B	C	D	$A \text{ AND } B$	$\text{NOT } C$	$\text{NOT } C \text{ AND } D$	$(A \text{ AND } B) \text{ OR } (\text{NOT } C \text{ AND } D)$
F	F	F	F	F	T	F	F
F	F	F	T	F	T	T	T
F	T	F	F	F	T	F	F
F	T	F	T	F	T	T	T
T	F	F	F	F	T	F	F
T	F	F	T	F	T	T	T
T	T	F	F	T	T	F	T
T	T	F	T	T	T	T	T
F	F	T	F	F	F	F	F
F	F	T	T	F	F	F	F
F	T	T	F	F	F	F	F
F	T	T	T	F	F	F	F
T	F	T	F	F	F	F	F
T	F	T	T	F	F	F	F
T	T	T	F	T	F	F	T
T	T	T	T	T	F	F	T

Question 2

$$(n)(n+1)(n+2)(n+3) = 24k, \quad k \in \mathbb{Z}, n \in \mathbb{Z},$$

Proof

$$P = (n)(n+1)(n+2)(n+3)$$

Since we are looking at the products of four consecutive integers we know that two are divisible by two, one of those two also divisible by four and another by three

Thus P is divisible by 2,3 and 4. If P is divisible by 2,3 and 4 it must also be divisible by $2 \times 3 \times 4 = 24$

Question 3

Proof by contrapositive

Trying to prove $p \rightarrow q$

Contrapositive is $\text{NOT } q \rightarrow \text{NOT } p$

Restatement of Question

$2^n - 1$ is prime $\rightarrow n$ is prime

n is NOT prime $\rightarrow 2^n - 1$ is NOT prime

Proof

$n \in \mathbb{N}$

Assume n is not prime and therefore can be written as a product of two integers, where $1 < a, b < n$

$$n = ab$$

$$2^n - 1 = 2^{ab} - 1$$

Then using the expression $x^k - 1 = (x - 1)((x)^{k-1} + (x)^{k-2} \dots x + 1)$, we get

$$2^n - 1 = (2^a - 1)((2^a)^{b-1} + (2^a)^{b-2} \dots 2^a + 1)$$

Since $a, b > 0$, we have that $2^n - 1$ can be written as the product of two integers, meaning that if n is not a prime then $2^n - 1$ is also not a prime.

Then the original statement, if $2^n - 1$ is prime then n is prime must be true

Question 4

For any integers m and n where $n \neq 0$, $m + \sqrt{2}n$ is not rational

Step 1

Assume $n\sqrt{2}$ is rational,

$n\sqrt{2} = \frac{a}{b}$ where the fraction is in its simplest form, i.e. highest common factor of a and b is 1 and $b \neq 0$, $a, b \in \mathbb{Z}$

$$\begin{aligned} 2n^2 &= \frac{a^2}{b^2} \\ 2(b^2n^2) &= a^2 \end{aligned}$$

Thus a^2 and a are even

$$a = 2k, \quad z \in \mathbb{Z}$$

$$\begin{aligned} 2(b^2n^2) &= 4k^2 \\ n^2b^2 &= 2k^2 \end{aligned}$$

Thus n^2b^2 is even, and at least one of the factors must be even

If n^2 is odd then b^2 and b must be even, which contradicts the assumption that the highest common factor of a, b is 1, a contradiction

If n^2 is even then n is even, then looking back at the original equation $n\sqrt{2} = \frac{a}{b}$ we see that n and a are even, thus

$$p\sqrt{2} = \frac{k}{b}$$

This is a smaller fraction which represents the same number, a contradiction

Thus $n\sqrt{2}$ must be irrational

Step 2

We now will check the case $m + \sqrt{2}n$, knowing $n\sqrt{2}$ is irrational

Assume $m + \sqrt{2}n$ is rational,

$m + n\sqrt{2} = \frac{a}{b}$ where the fraction is in its simplest form, i.e. highest common factor of a and b is 1 and $b \neq 0$, $a, b \in \mathbb{Z}$

We can reorder the equation to get

$$\begin{aligned} n\sqrt{2} &= \frac{a}{b} - m \\ n\sqrt{2} &= \frac{a - mb}{b} \end{aligned}$$

This shows that $n\sqrt{2}$ can be written as a fraction, which we know is false since we just proved it's irrational. This is a contradiction, thus $m + n\sqrt{2}$ is also irrational

Question 5

$$\sum_{i=0}^n i^2 = \frac{1}{6}n(n+1)(2n+1)$$

Proof by Induction

Check base case ($n = 0$)

$$0 = \frac{1}{6}(0)(1)(2) = 0$$

Assume formula holds for $n = k$

$$\sum_{i=0}^k i^2 = \frac{1}{6}k(k+1)(2k+1)$$

$$\sum_{i=0}^k i^2 + (k+1)^2 = \frac{1}{6}k(k+1)(2k+1) + (k+1)^2$$

$$\sum_{i=0}^{k+1} i^2 = \frac{1}{6}(2k^3 + 3k^2 + k + 6k^2 + 12k + 6)$$

$$= \frac{1}{6}(2k^3 + 9k^2 + 13k + 6)$$

$$= \frac{1}{6}(k+1)(2k^2 + 7k + 6)$$

$$= \frac{1}{6}(k+1)(k+2)(2k+3)$$

$$= \frac{1}{6}(k+1)((k+1)+1)(2(k+1)+1)$$

Thus the formula holds in the induction step, and the result is proven